AN ETALE COVER OF THE TARGET INDUCES AN ETALE REPRESENTABLE MORPHISM OF MOR FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Section functor algebraicity. lemma-space-of-sections (same chapter, named). Provides the etale algebraic space F of sections.
- Surjectivity of sections. lemma-surjection-space-of-sections (same chapter, named). Gives surjectivity when X' -> X is surjective.

Source reference. Original lemma 05XR: sources/stacks-criteria/pinned-original-source.tex, source lines 1379–1379. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05XQ: sources/stacks-criteria/pinned-original-source.tex, source lines 1321–1321. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $Z \to B$ and $X' \to X \to B$ be morphisms of algebraic spaces over S. Assume*

(1) *$X' \to X$ is étale, and*
(2) *$Z \to B$ is finite locally free.*

Then $Mor_B(Z, X') \to Mor_B(Z, X)$ is representable by algebraic spaces and étale. If $X' \to X$ is also surjective, then $Mor_B(Z, X') \to Mor_B(Z, X)$ is surjective.

Proof. Let U be a scheme and let $\xi = (a, b)$ be an element of $Mor_B(Z, X)(U)$. We have to prove that the functor

$$h_U \times_{\xi, Mor_B(Z,X)} Mor_B(Z, X')$$

is representable by an algebraic space étale over U. Set $Z_U = U \times_{a,B} Z$ and $W = Z_U \times_{b,X} X'$. Then $W \to Z_U \to U$ is as in Lemma 05XR and the sheaf F defined there is identified with the fibre product displayed above. Hence the first assertion of the lemma. The second assertion follows from this and Lemma 05XQ which guarantees that $F \to U$ is surjective in the situation above. □

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